

The ALU was designed using hierarchical decomposition: alu1 for 1-bit operations, alu2 for 2-bit combination, and alu4 for full 4-bit ripple-carry behavior. This modular design improves clarity, reuse, and debugging. The ALU supports AND, OR, ADD, SUB, ROL, ROR, and SLT operations. Subtraction is implemented using Binvert logic, which automatically inverts B when S=011. The 4-bit ALU includes Zero, Carry-Out, and Overflow flags for full CPU-style behavior.

A structured testbench was used, sweeping values of A, B, and operation S. Results were validated using GTKWave. All operations behaved correctly:

- Logical operations produced expected bitwise outputs.
- ADD and SUB behaved correctly with proper carry propagation.
- Overflow triggered only on valid signed overflow conditions.
- Zero flag activated correctly when R=0000.

4. Conclusion

The ALU performed correctly in all tested conditions. The modular structure, ripple-carry chaining, and clean operation logic resulted in a correct and maintainable design.